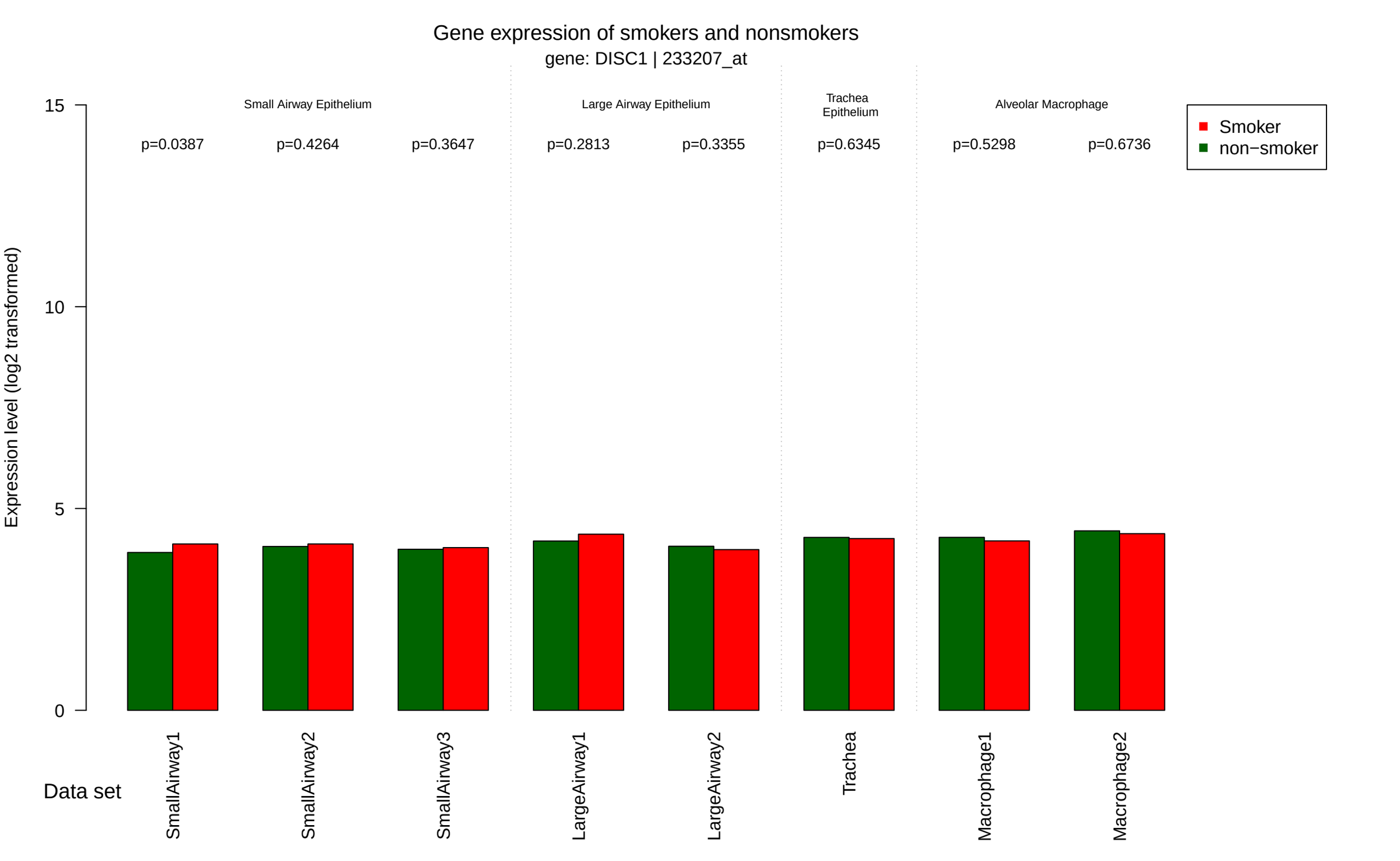
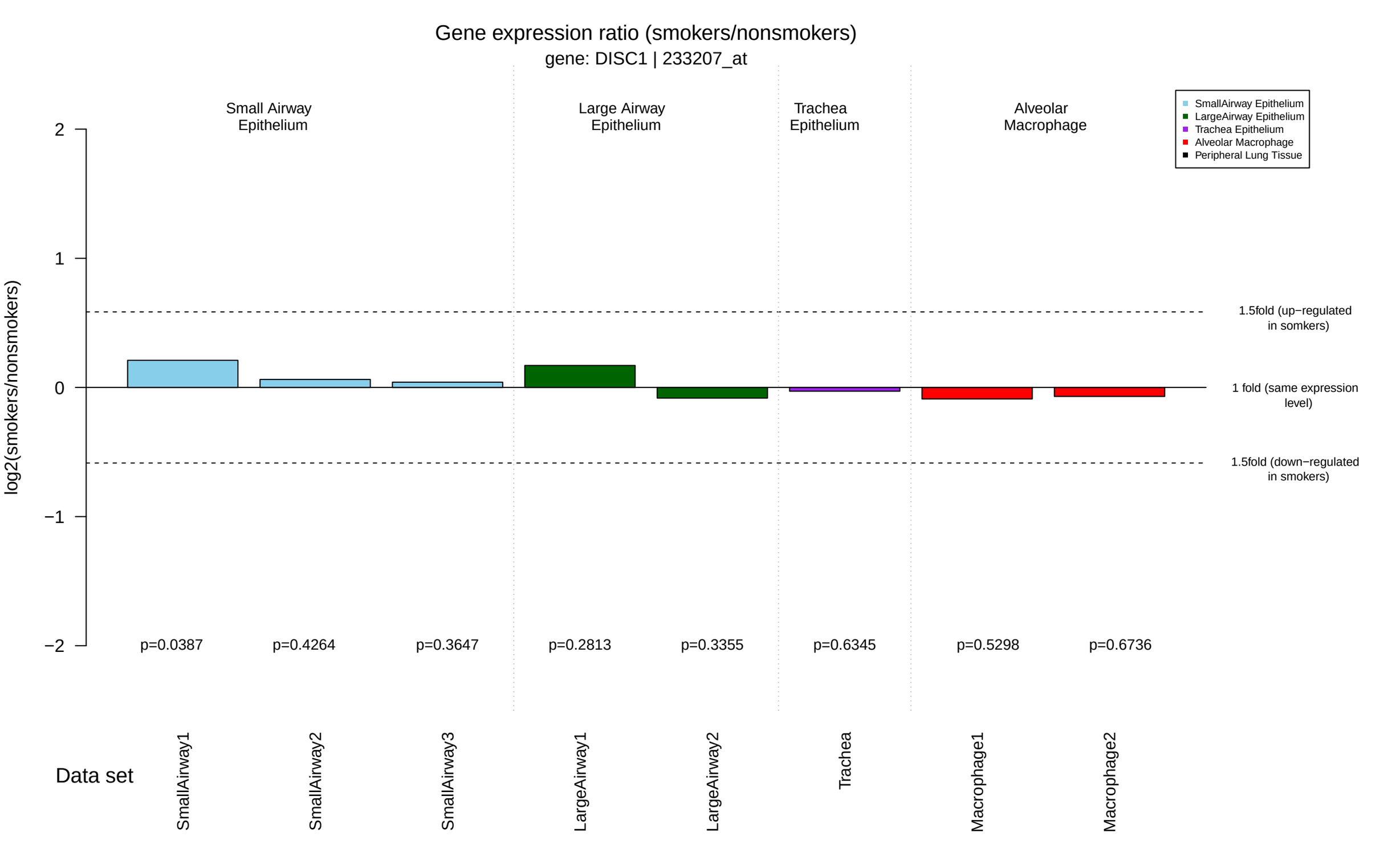
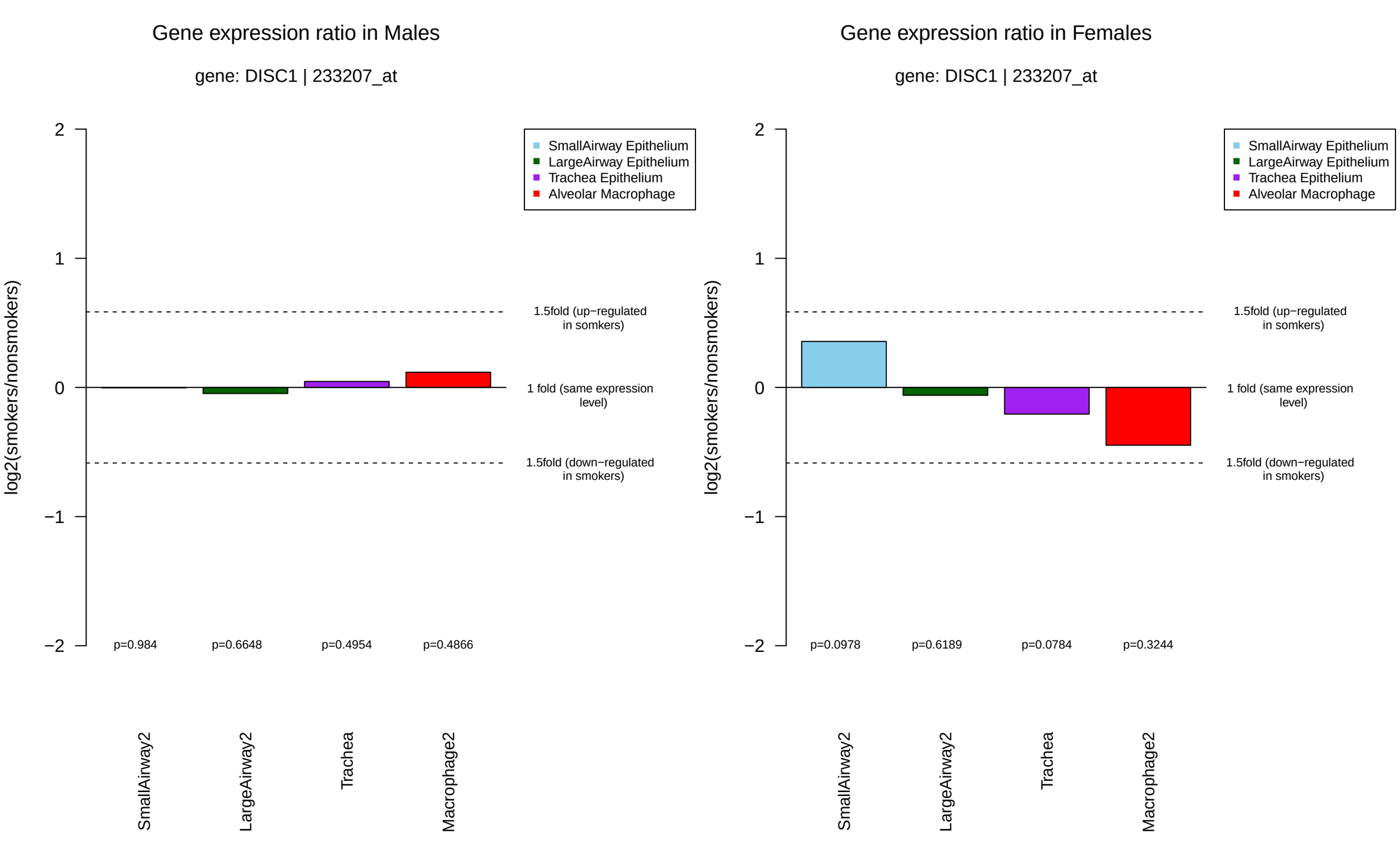
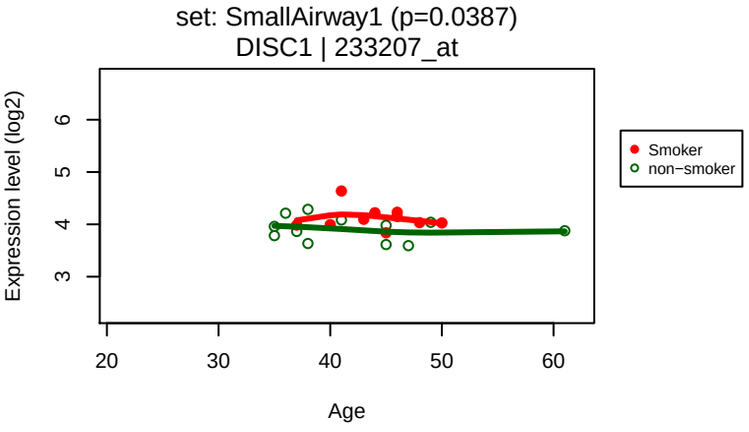
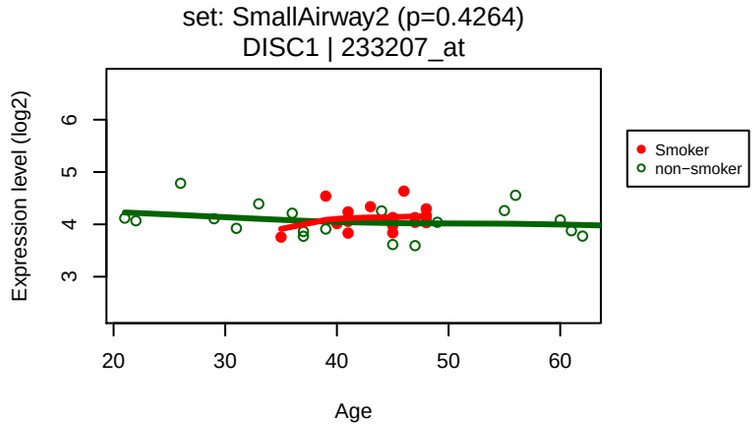
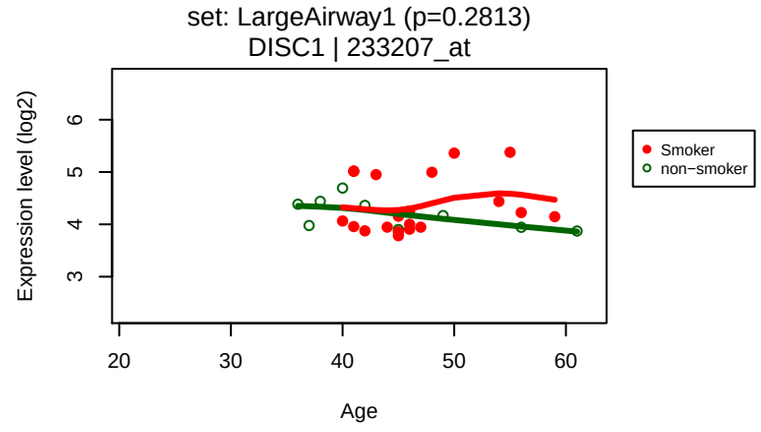

Gene expression of smokers and nonsmokers
gene: DISC1 | 233207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0387
p=0.4264
p=0.3647
p=0.2813
p=0.3355
p=0.6345
p=0.5298
p=0.6736
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

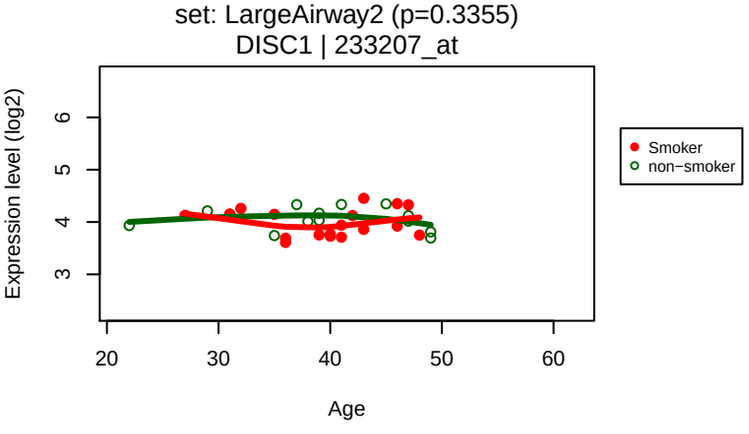

Gene expression ratio (smokers/nonsmokers)
gene: DISC1 | 233207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0387
p=0.4264
p=0.3647
p=0.2813
p=0.3355
p=0.6345
p=0.5298
p=0.6736
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

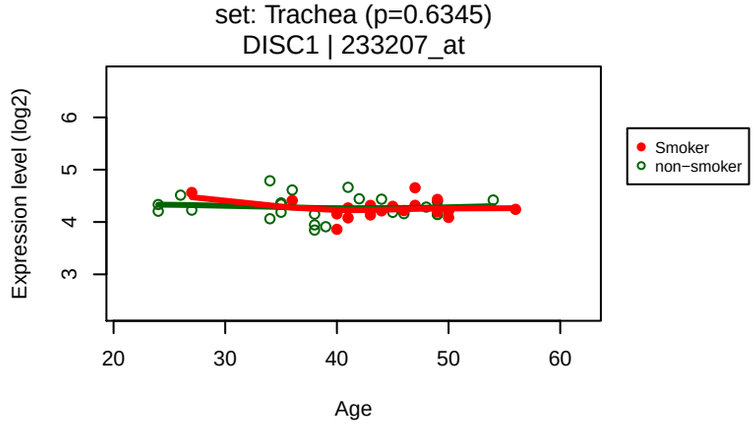

Gene expression ratio in Males
gene: DISC1 | 233207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.984
p=0.6648
p=0.4954
p=0.4866
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DISC1 | 233207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0978
p=0.6189
p=0.0784
p=0.3244
SmallAirway2
LargeAirway2
Trachea
Macrophage2

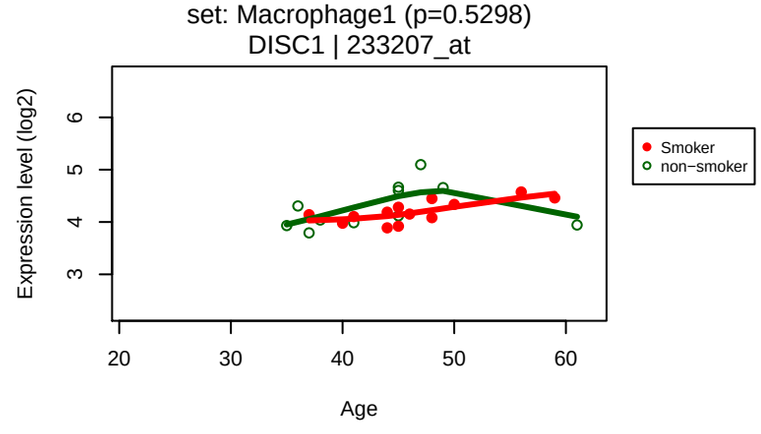

set: SmallAirway1 (p=0.0387)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker

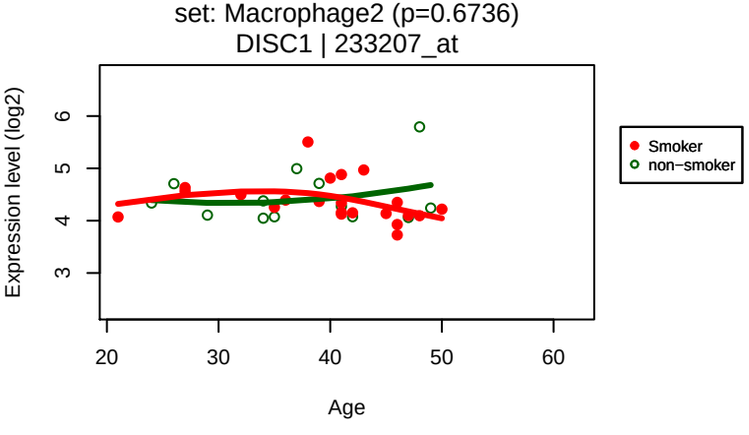

set: SmallAirway2 (p=0.4264)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker

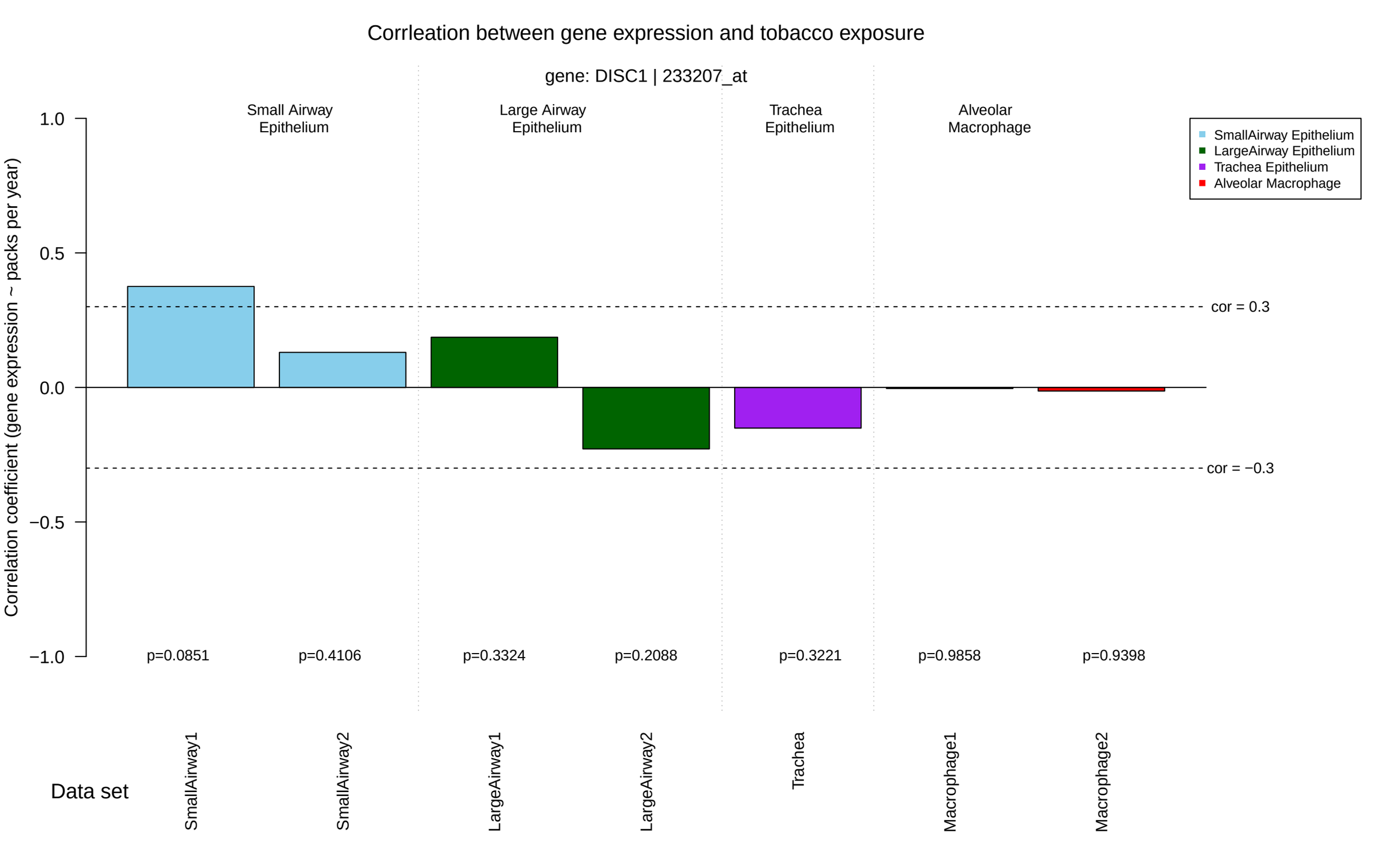

set: LargeAirway1 (p=0.2813)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3355)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6345)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5298)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6736)
DISC1 | 233207_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DISC1 | 233207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0851
p=0.4106
p=0.3324
p=0.2088
p=0.3221
p=0.9858
p=0.9398
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2